Python Activity-Based Learning:

Instructions: Select any 10 case studies and develop complete Python solutions using only the assigned syllabus: computational thinking, algorithms, flowcharts, pseudocode, variables, data types, operators, conditional statements, loops, strings, pattern generation, lists, nested lists, tuples, tuple operations, and unpacking.

1. University Scholarship Allocation: Design a system that evaluates applicants using aggregate marks, subject-wise minimum scores, attendance, and income criteria, ranks eligible students, and resolves ties using predefined rules.

2. Progressive Electricity Billing: Calculate household bills using multiple consumption slabs, fixed charges, surcharges, and rebates, then compare bills across several months and identify unusual increases.

3. Hospital Billing and Patient Classification: Process consultation, diagnostic, and treatment charges, apply conditional concessions, validate billing constraints, and classify patients according to a supplied payment policy.

4. Employee Payroll and Deduction Analysis: Calculate gross salary from basic pay, allowances, overtime, and incentives; apply progressive deductions and generate a formatted salary statement.

5. Multi-Plan Telecommunications Cost Optimization: Evaluate several mobile plans against usage, validity, and budget constraints; calculate excess charges and identify the most economical eligible plan.

6. Railway Fare and Reservation Calculator: Calculate fares for multiple passengers using distance slabs, travel classes, age-based concessions, reservation fees, and specified cancellation penalties.

7. Retail Billing with Multiple Discount Rules: Process multiple products, quantities, category-specific taxes, threshold discounts, and maximum concession limits to generate an itemized invoice.

8. Manufacturing Cost and Profitability Analysis: Calculate material costs, labour charges, wastage, machine costs, and selling prices; identify whether production meets a specified profit target.

9. Multi-Currency Travel Budget Analysis: Convert travel expenses using supplied exchange rates, incorporate conversion fees, compare planned spending against a budget, and identify the largest expense category.

10. Campus Event Financial Planning: Calculate venue, equipment, catering, transport, and certificate costs; determine participant contributions, sponsorship shortfalls, and the minimum registration count needed to break even.

11. Examination Result and Grace-Mark Validation: Process marks across multiple subjects, apply subject-wise pass criteria and a specified grace-mark policy, and generate final classifications without exceeding maximum marks.

12. Attendance Recovery Computation: Calculate current attendance percentages and determine the minimum consecutive classes required to reach an eligibility threshold under different future attendance scenarios.

13. ATM Transaction Sequence Simulation: Process a sequence of deposits and withdrawals, maintain a simulated balance, enforce withdrawal limits, reject invalid transactions, and produce a transaction summary.

14. Warehouse Inventory Reconciliation: Compare opening stock, incoming quantities, dispatched quantities, and closing stock across multiple products; detect discrepancies and calculate replenishment requirements.

15. Smart Parking Fee Management: Calculate charges from entry and exit times, vehicle categories, overnight parking, daily limits, and penalty conditions while validating time inputs.

16. Multi-Day Water Tank Simulation: Process changing inflow and outflow quantities, update water levels, detect overflow and shortages, and determine the first interval in which the tank reaches a specified threshold.

17. Student Performance Trend Analysis: Compare assessment scores across multiple examinations, calculate improvements and declines, identify consistently weak subjects, and classify students according to defined criteria.

18. Fuel Consumption and Route Comparison: Calculate fuel requirements and journey costs for several vehicles using distance, mileage, fuel prices, and reserve requirements; identify the lowest-cost feasible option.

19. Multi-Product Shopping Budget Constraint: Evaluate a shopping basket containing multiple products and quantities, apply discount conditions, and determine which purchases must be reduced to remain within a fixed budget.

20. University Admission Decision Engine: Evaluate candidates using subject eligibility, aggregate marks, course-specific thresholds, and tie-breaking criteria; generate a ranked list of eligible applicants.

21. Traffic Violation Penalty Calculator: Process multiple violations using vehicle category, offence type, repeat-offence status, and predefined fine rules; calculate the final penalty and identify disqualification conditions.

22. Laboratory Chemical Inventory Validation: Analyse hypothetical chemical inventory records using storage categories, maximum quantities, and compatibility restrictions supplied in the problem; flag invalid combinations.

23. Student Timetable Conflict Detection: Compare course schedules represented by strings and tuples, detect overlapping periods, identify unavailable rooms, and validate a proposed timetable.

24. Courier Charge Optimization: Calculate delivery charges based on parcel weight, dimensions, distance zones, urgency, and declared restrictions; compare service options and select an eligible one.

25. Progressive Water Tariff Analysis: Calculate bills for multiple households using consumption slabs, fixed fees, and penalties; compare consumption patterns and identify excessive usage.

26. Nested Loop Examination Seating Generator: Generate a seating arrangement with row and column identifiers, reserved positions, alternate seating restrictions, and fixed classroom capacities.

27. Advanced Numeric Pattern Generator: Produce a hollow diamond or symmetric numerical pattern based on user-specified dimensions, with validation for invalid dimensions and carefully aligned output.

28. Repeated Transaction Detection: Process transaction identifiers and amounts in a list, detect duplicate transactions, identify threshold violations, and generate a list of suspicious entries.

29. Password Policy Validation Engine: Evaluate sample password strings against minimum length, character-category, repetition, and prohibited-substring rules; explain each failed condition.

30. Student Email and Identifier Validation: Validate student email-like strings and registration identifiers against supplied formatting rules, reporting malformed separators, invalid character groups, and missing components.

31. String-Based Examination Answer Evaluation: Compare answer strings against an answer key, calculate correct, incorrect, and unanswered responses, apply negative marking, and produce student-wise results.

32. Text-Based Invoice Data Extraction: Process formatted invoice strings using basic string operations, extract product codes, quantities, and amounts, and flag malformed records.

33. Campus Registration Identifier Analyzer: Examine student identification strings, validate their structure, extract specified components, and classify records according to branch and admission year codes.

34. Student Feedback Sentiment Scoring: Analyse multiple feedback sentences using a supplied keyword list, count positive and negative indicators, calculate scores, and classify feedback using predefined thresholds.

35. Text Compression Using Character Runs: Encode a string by grouping consecutive repeated characters and compare original and encoded lengths, including cases where encoding is not beneficial.

36. Repeated Word and Phrase Detection: Process a paragraph, normalize letter case, count word occurrences using permitted string and list operations, and identify repeated words and their positions.

37. Nested Loop Calendar Generator: Generate a formatted monthly calendar using the supplied starting weekday and number of days, correctly aligning weeks and handling valid month lengths.

38. ATM Menu-Driven Banking Simulation: Implement repeated menu operations for deposits, withdrawals, balance inquiries, and transaction summaries, with validation and specified account limits.

39. Multi-Stage Examination Eligibility System: Determine eligibility for a final examination using attendance, laboratory completion, internal assessment, and fee-clearance conditions, reporting every unmet requirement.

40. Traffic Flow and Congestion Analysis: Process vehicle counts across multiple roads and time intervals, calculate peak traffic periods, classify congestion levels, and identify roads exceeding capacity thresholds.

41. Nested List Classroom Occupancy Analysis: Represent multiple classrooms and seating positions as nested lists; allocate students, preserve reserved seats, detect duplicate assignments, and calculate vacant capacity.

42. Student Marks Matrix Analysis: Analyse a matrix of student marks across subjects, calculate row-wise and column-wise statistics, identify top performers, and detect subjects with low average performance.

43. Multi-Branch Sales Matrix Comparison: Store product sales across branches and days in nested lists, calculate product totals, compare branches, and identify the best-performing product for each branch.

44. Hospital Ward Occupancy Monitoring: Analyse a nested list of bed occupancy counts by ward and day, calculate utilization percentages, detect capacity violations, and identify peak-demand periods.

45. Weekly Electricity Consumption Matrix: Process appliance-wise electricity readings across seven days, calculate daily and appliance totals, and identify consumption exceeding specified limits.

46. Inventory Matrix Reconciliation: Compare product quantities across warehouses using nested lists, identify shortages and surpluses, and calculate the transfers required to meet specified stock targets.

47. Sports Tournament Ranking System: Store match results and team scores, calculate total points and tie-break values, and produce rankings under a specified competition policy.

48. Tuple-Based Employee Shift Conflict Detection: Represent employee shifts as tuples, compare time intervals, detect overlapping assignments, and identify employees with conflicting schedules.

49. Tuple-Based Product Catalogue Audit: Store product identifiers, prices, and stock quantities in tuples; detect invalid records, calculate inventory values, and identify items requiring replenishment.

50. Tuple-Based Route Distance Analysis: Store geographical coordinates as tuples, calculate distances between successive points using a supplied formula, and identify the longest route segment.

51. Bus Passenger Occupancy Simulation: Process boarding and alighting counts across multiple stops, calculate passenger counts at each stop, and flag any capacity violations.

52. Hackathon Team Formation and Ranking: Analyse student scores and team compositions, validate team sizes, aggregate round-wise scores, and resolve tied rankings using predefined rules.

53. Food Waste Monitoring System: Compare prepared meals, meals served, and leftovers across several days; calculate wastage percentages and identify days exceeding acceptable limits.

54. Multi-Branch Expense Matrix Analysis: Analyse nested lists containing monthly expenses across branches and categories, calculate totals and averages, and identify the largest contributors to expenditure.

55. Tuple-Based Transaction Audit: Process transaction tuples containing identifiers, amounts, and transaction types; detect duplicate identifiers, invalid amounts, and suspicious sequences.

56. Student Group Formation with Score Balancing: Arrange students using a list of scores and allocate them into groups of specified sizes while minimizing differences between group totals under defined allocation rules.

57. Smart Canteen Order and Stock Management: Process successive customer orders, calculate bills, update available inventory, reject unavailable quantities, and summarize daily sales and remaining stock.

58. Library Borrowing and Fine Reconciliation: Analyse tuples representing book identifiers, borrowing periods, and return delays; calculate fines, identify overdue records, and reconcile book availability.

59. Campus Transport Allocation: Determine the number of buses needed for several routes, account for bus capacities and passenger demand, and identify unserved passengers.

60. Manufacturing Quality Control Matrix: Analyse production quantities and defective units across products and batches, calculate defect percentages, and identify batches exceeding quality thresholds.

61. Nested List Examination Seating Validation: Validate a complete seating matrix for duplicate roll numbers, reserved positions, empty seats, and invalid student allocations.

62. Water Distribution Fairness Analysis: Analyse area-wise water supply across several days, calculate shortages against demand, compare distribution levels, and identify the most underserved areas.

63. Multi-Product Price Fluctuation Analysis: Process historical product prices stored in lists, calculate percentage changes, identify maximum fluctuations, and flag products exceeding a supplied threshold.

64. Tuple-Based Course Registration Validation: Represent course registrations as tuples, identify duplicate registrations and invalid course combinations, and determine students whose selections violate supplied rules.

65. Campus Resource Allocation Under Constraints: Compare equipment requirements with available resources across multiple activities, calculate shortages, and produce a feasible allocation using a specified priority order.

66. Student Academic Risk Classification: Combine lists of marks, attendance, and laboratory scores to classify students into risk categories, identify multiple simultaneous risk factors, and generate a summary.

67. Retail Order Dispatch and Return Analysis: Process order quantities, dispatch records, and returned quantities; calculate outstanding orders, update inventory, and detect inconsistent records.

68. Energy Conservation Before-and-After Analysis: Compare appliance-wise electricity usage before and after an intervention, calculate savings percentages, and rank appliances by achieved savings.

69. Nested List Sports Performance Analysis: Analyse players' scores across multiple matches, calculate consistency measures, identify performance trends, and rank players under supplied tie-breaking rules.

70. Multi-Stage Scholarship Shortlisting: Apply minimum academic eligibility, attendance requirements, income conditions, and ranking rules to select candidates within a fixed scholarship quota.

71. Integrated University Fee and Eligibility Audit: Combine fee payments, attendance, internal marks, and examination eligibility records; identify students with outstanding conditions and generate a reason-wise report.

72. Smart Hostel Resource Monitoring: Analyse room-wise electricity and water consumption, calculate charges, identify threshold violations, and compare resource utilization across floors.

73. Multi-Route Courier Dispatch Planning: Process package weights, destination zones, vehicle capacities, and delivery priorities; allocate packages to vehicles and identify packages that cannot be dispatched.

74. Campus Festival Registration and Financial Audit: Reconcile participant lists, registration fees, sponsorship contributions, and expenses; detect duplicate registrations and calculate the final financial balance.

75. Integrated Student Record Consistency Checker: Validate student identifiers, names, marks, attendance, and course records stored in lists and tuples; detect duplicates, missing values, and inconsistent academic records.

76. University Examination Result Moderation: Process marks across multiple subjects, apply specified moderation rules, preserve maximum-mark constraints, and compare original and moderated outcomes.

77. Smart Electricity Billing for Multiple Apartments: Calculate apartment-wise bills using progressive tariffs, fixed charges, and rebates; identify billing discrepancies and rank apartments by consumption.

78. Hospital Resource Allocation and Capacity Monitoring: Analyse nested lists of bed occupancy, available staff, and equipment across wards; identify resource shortages and prioritize cases using supplied rules.

79. Integrated Warehouse Order Prioritization: Process product demand, stock levels, order quantities, and priority categories; determine fulfilment feasibility and calculate outstanding quantities for each order.

80. Multi-Day Bus Route Performance Analysis: Analyse passenger demand, journey times, fuel usage, and bus capacities; identify overloaded routes and compare operating efficiency using supplied criteria.

81. Campus Event Seating and Entry Validation: Generate seating assignments from nested lists, validate tickets against registration identifiers, detect duplicate entries, and report available seating capacity.

82. Multi-Criteria Internship Candidate Ranking: Evaluate candidate marks, assessment scores, attendance, and eligibility conditions; rank qualified candidates and apply deterministic tie-breaking rules.

83. Food Supply Chain Stock Reconciliation: Compare quantities received, distributed, damaged, and remaining across several locations; identify discrepancies and calculate replenishment requirements.

84. Environmental Monitoring Data Analyzer: Analyse nested lists of air-quality or water-quality readings, calculate averages and extremes, and classify observations using supplied threshold values.

85. Smart Classroom Timetable Allocation: Represent courses, rooms, and periods using lists and tuples; validate classroom capacities, detect clashes, and check whether all courses can be accommodated under the stated constraints.

86. Multi-Stage ATM Transaction Audit: Process a sequence of deposits, withdrawals, and transfers in a simulated account, maintain balances, detect invalid operations, and report suspicious repeated patterns.

87. Examination Question Difficulty and Performance Analysis: Compare student answer strings against a key, calculate question-wise success rates, identify the most frequently missed questions, and classify question performance.

88. Multi-Branch Retail Inventory Transfer: Compare branch-level stock lists with target stock levels, calculate surpluses and shortages, and generate transfers while respecting supplied transfer constraints.

89. University Hostel Room Allocation: Allocate students to rooms according to capacity, room eligibility, and specified grouping constraints; detect duplicate assignments and calculate unallocated demand.

90. Integrated Campus Energy and Water Audit: Analyse consumption across buildings and days, calculate costs and percentage changes, identify excessive consumption, and rank buildings using a specified scoring method.

91. Advanced String-Based Registration Record Analyzer: Parse structured registration strings, validate field lengths and formats, extract specified components, detect duplicates, and produce error classifications.

92. Multi-Event Sports Scheduling and Scoring: Represent event schedules and team scores using lists and tuples, identify scheduling conflicts, aggregate points, and generate final rankings.

93. Smart Public Transport Fare Reconciliation: Calculate passenger fares across multiple routes and passenger categories, reconcile collected revenue with expected revenue, and identify discrepancies.

94. Disaster Relief Supply Allocation: Compare stock lists against area-wise requirements, prioritize urgent shortages, allocate limited supplies according to a prescribed policy, and calculate remaining unmet demand.

95. Integrated Manufacturing Production Planner: Analyse raw materials, production quantities, defect counts, and order demand; calculate production feasibility and identify shortages and quality violations.

96. Multi-Criteria Library Resource Distribution: Analyse book availability, borrowing demand, and branch-wise requirements; allocate copies according to supplied priority rules and calculate unmet demand.

97. Smart Campus Incident and Maintenance Prioritization: Classify hypothetical maintenance reports using severity, location, and reported frequency; rank cases using predefined rules and generate a priority summary.

98. University Academic Progression Audit: Combine semester marks, subject pass conditions, attendance, and prerequisite completion records to determine progression eligibility and identify outstanding requirements.

99. Integrated E-Commerce Order Processing and Reconciliation: Process product orders, discounts, dispatches, returns, and stock updates; calculate revenue, identify discrepancies, and generate a final inventory summary.

100. Comprehensive Smart Campus Management Simulation: Develop a menu-driven program that integrates student records, attendance, marks, event registration, resource allocation, and summary reporting using conditional statements, loops, strings, lists, nested lists, tuples, and tuple unpacking.

Mandatory deliverables for each of the 10 selected case studies: Problem statement, objectives, assumptions, algorithm, flowchart, pseudocode, Python source code, test cases, output screenshots, and conclusion. Upload all materials to the student's own GitHub repository named `Python-Activity-Based-Learning` and submit the repository link after the Dussehra holidays.

Evaluation: 10 marks, awarded by the respective Python Programming faculty member as per the regular academic schedule.